



ENHANCING HOST STRATEGY USING DATA ANALYSIS ON AIRBNB

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PROJECT OBJECTIVES

1

Assess Airbnb hosting strategies across diverse locations through data analysis

2

Uncover patterns in listing prices, occupancy rate, and the influence of amenities on revenue

3

Evaluate seasonality, local events, and neighborhood effects on demand and profitability

4

Equip Airbnb hosts with actionable strategies for optimizing pricing, maximizing occupancy, and enhancing guest experience

5

Offer data-driven guidance to investors on identifying high-demand neighborhoods for profitable investments

DATASET OVERVIEW



DATA SOURCE

Airbnb Open Data Inside Airbnb

<https://insideairbnb.com/get-the-data/>

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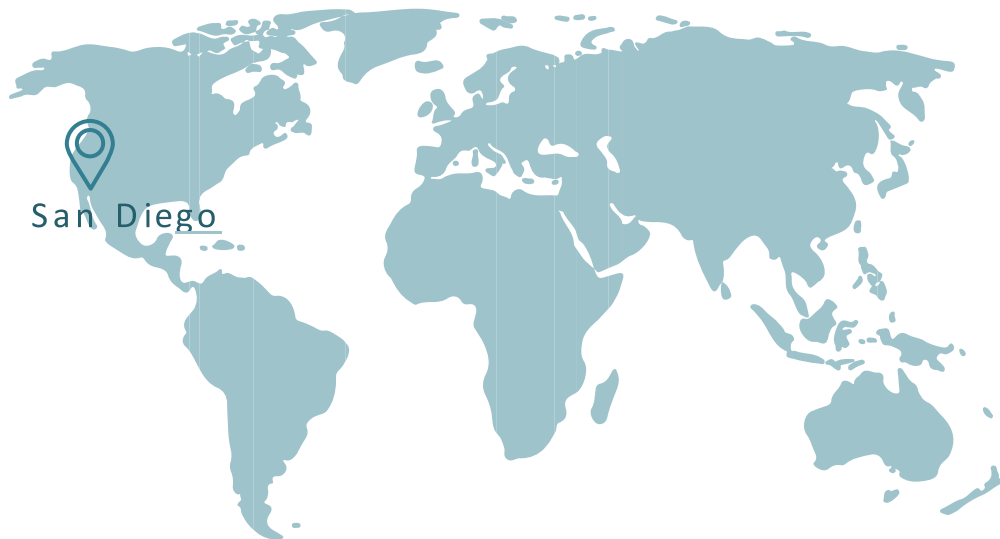


CITY

San Diego, California,

USA  DATASETS

- Listings
- Calendar
- Reviews



DATASETS



LISTINGS

- Contains detailed data on individual Airbnb listings
- Including attributes like price, number of bedrooms, and amenities



CALENDAR

- Provides daily availability and pricing information for each listing over a specified period



REVIEWS

- Includes reviews for each listing, detailing the date and reviewer information

LIBRARIES

01

`folium`

A library for creating

map using Python

`folium.plugins.MarkerClus`

A plugin in Folium for `ter` interactive maps

markers based on

clustering

02

`wordcloud`

A library for creating
visualizations from

A module for handling word cloud
Python text data

`proximity`
`datetime`

date and time in

04

`matplotlib.pyplot`

05

A plotting library for

creating static, A module in scikit-learn for animated, and converting a collection of text
interactive documents to a matrix

`sklearn.feature_extraction.t`

`ext.CountVectorizer`

visualizations in

of token counts

Python

06



CODE DEMONSTRATION KEY INSIGHTS

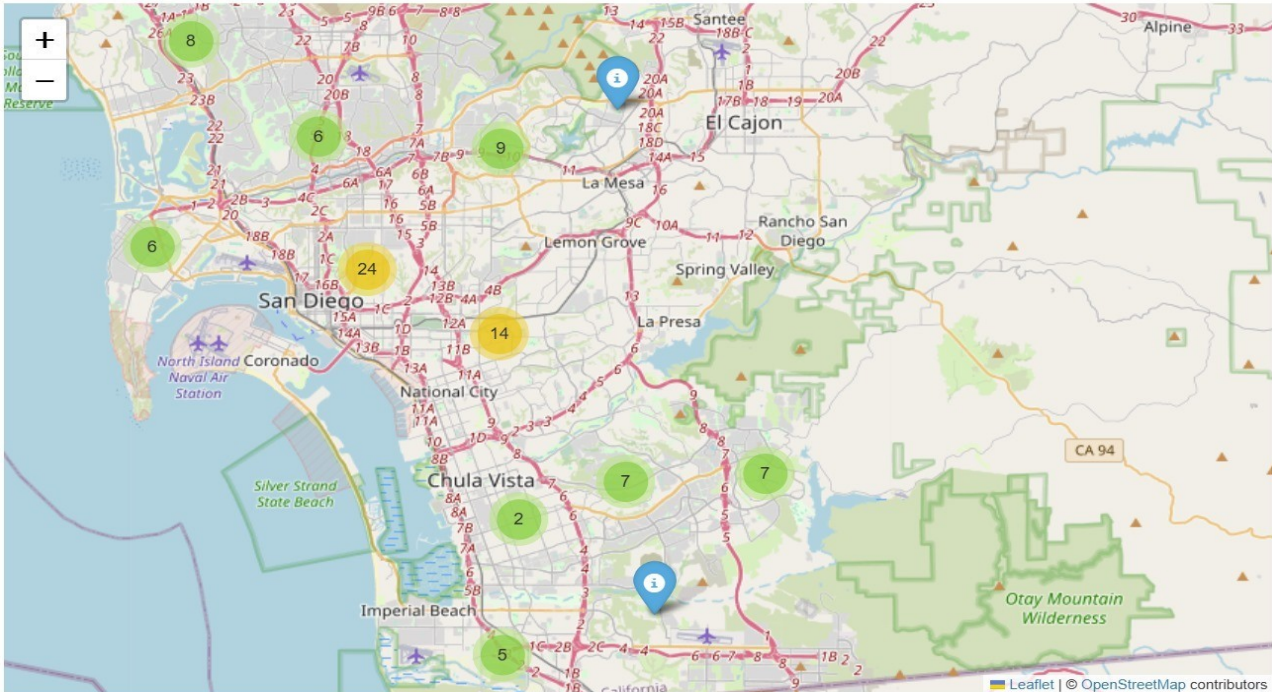


REVIEW SCORES AND POPULARITY



Q2

NEIGHBORHOOD INFLUENCE ON PRICES AND AVAILABILITY



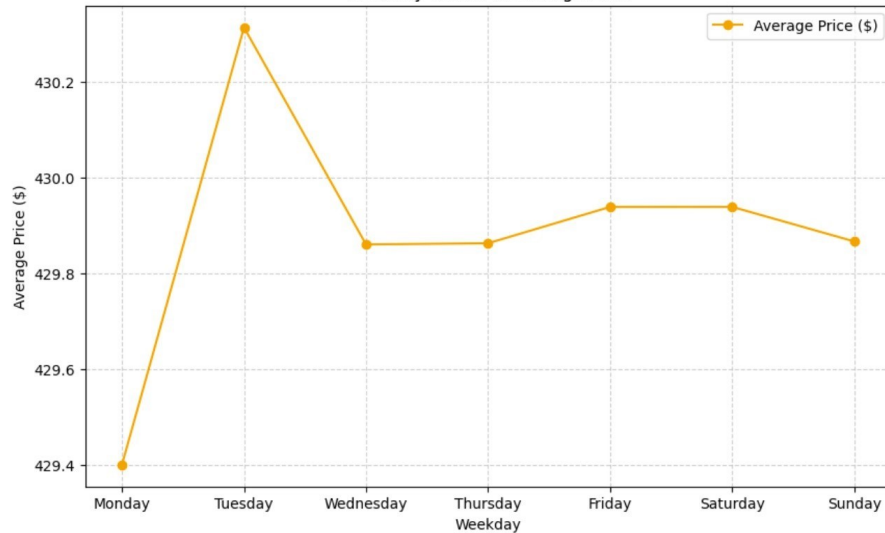
Q3

SEASONAL FLUCTUATIONS IN LISTING PRICES

Monthly Trends in Average Price

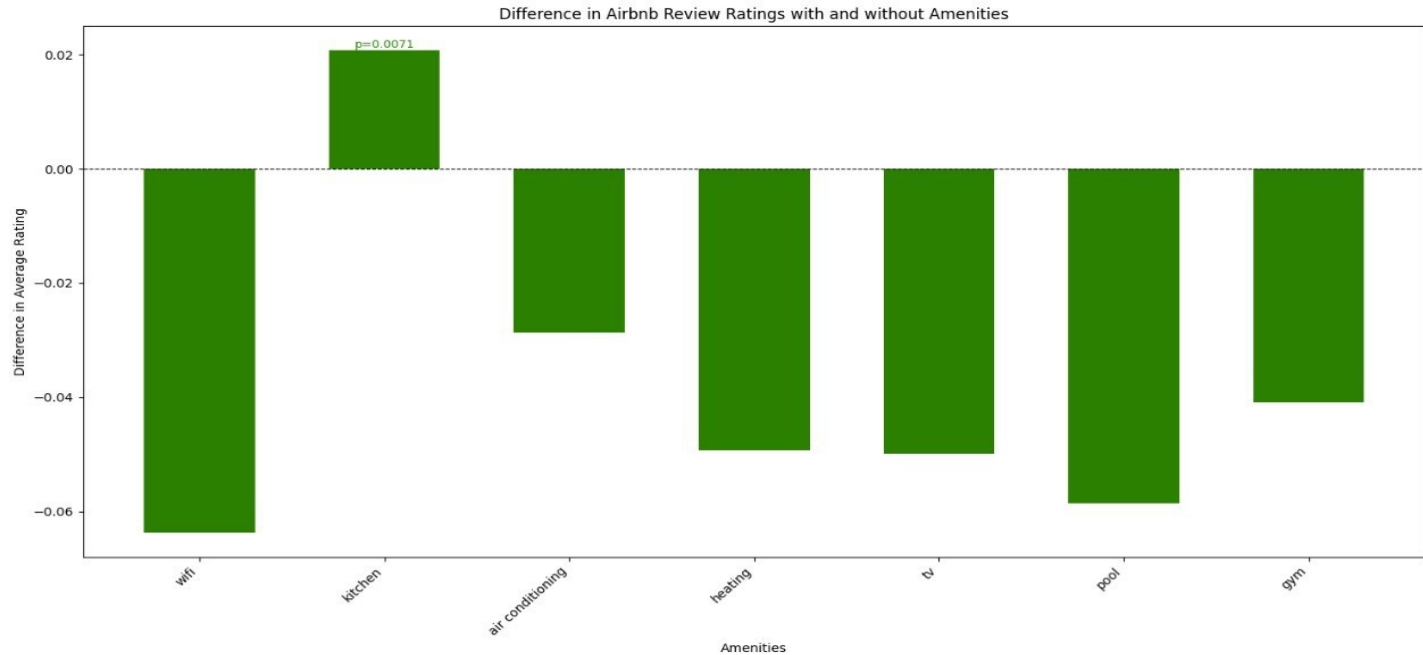


Weekday Trends in Average Price



Q4

EFFECT OF AMENITIES ON AVERAGE RATINGS





IMPACT OF LOCAL HOSTS ON

BOOKING RATES AND REVIEW SCORES

Booking Rate - Local Hosts: 47.60%

Booking Rate - Non-Local Hosts: 41.73%

Average Review Score - Local Hosts: 4.83

Average Review Score - Non-Local Hosts: 4.78

CONCLUSION

- **Optimized Hosting Strategies:** Data insights help hosts enhance guest experiences and maximize revenue.

- **Actionable Investment Guidance:** Identified high-demand neighborhoods and key profitability factors for investors.
- **Focus on Key Factors:** Analyzed pricing, occupancy, amenities, seasonality, and neighborhood dynamics.
- **Competitive Edge:** Equipped stakeholders with strategies to thrive in the dynamic short-term rental market.
- **Significance of Data-Driven Decisions:** Demonstrated the critical role of analytics in fostering success and superior guest experiences.

THANK YOU